

Nikhilesh Vijay Chirutala

AI/ML Engineer

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SUMMARY

AI/ML Engineer with nearly 3 years of software engineering experience building production applications across AI, cloud, CRM, trading, and contact center platforms. Strong hands-on experience with Python, FastAPI, data processing, REST APIs, AWS, MongoDB, RAG, LLM integration, and cloud application development. Experienced in integrating LLM-based solutions into application workflows, developing data-processing and automation pipelines, troubleshooting production systems, and delivering reliable backend services. Familiar with machine learning workflows including data cleaning, feature preparation, model evaluation, and retraining, with a strong foundation in algorithms, data structures, and software design principles.

TECHNICAL SKILLS

AI & Machine Learning: RAG, LLM Integration, Machine Learning, Data Processing, Data Cleaning, Model Evaluation, Model Retraining, Feature Engineering, Vector Search, Prompt Engineering

Python & Data: Python, Pandas, NumPy, SQL, Data Wrangling, Data Analysis

ML & Algorithms: Classification, Regression, Clustering, Model Evaluation, Algorithms & Data Structures, OOP, Software Design Principles

Backend: FastAPI, REST APIs, API Development, Backend Automation, Authentication, RBAC, ACL

Frontend: React.js, Next.js, Angular, JavaScript (ES6+), TypeScript, Tailwind CSS, Bootstrap, Jinja2

Databases: MongoDB, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs, Testing & Tools: Twilio, GitHub API, Postman, Unit Testing, Jest, Git, GitHub

Professional: Technical Communication, Written & Verbal Communication, Cross-Functional Collaboration, Problem Solving

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

- Develop backend services and automation for a cloud contact center platform, supporting application workflows, data processing, and cloud-based operations.
- Design and maintain Python and FastAPI services and REST APIs, translating business requirements into backend functionality and integrations.
- Develop AWS-based data and storage workflows using Amazon S3, with validation, error handling, and consistency checks for production data.
- Investigate production issues, perform root-cause analysis, implement bug fixes and feature enhancements, and validate changes before release.

Key Projects:

CONVOSO — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, REST APIs

- Developed backend services supporting lead routing, call recording management, data processing, and cloud storage workflows.
- Designed recording-recovery workflows processing 10,000+ recordings per month and restoring missing or inconsistent recording data.
- Implemented S3 storage validation and metadata synchronization to maintain consistency between application records and cloud objects.
- Built automation to identify, recover, organize, and synchronize recordings across storage systems, reducing manual intervention.
- Added validation and error handling for failed recording operations and unexpected storage states to improve recovery reliability.
- Traced production pipeline failures across backend services, cloud storage, and metadata workflows to identify root causes and implement fixes.
- Monitored processing outcomes and validated production changes to maintain reliable data and storage operations.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Worked across four production applications, delivering backend, frontend, API integration, and application features across the stack.
- Built backend modules, REST APIs, and business logic using Python, FastAPI, and MongoDB for application-specific workflows and data processing.
- Developed frontend interfaces using React.js, Next.js, Angular, JavaScript, and TypeScript, integrating application workflows with APIs.
- Implemented authentication, RBAC, and ACL-based authorization to enforce access controls across application resources.
- Integrated third-party services including Twilio and GitHub API and supported application deployments across AWS and Vercel.
- Maintained GitHub Actions CI/CD workflows and handled testing, debugging, bug fixes, and production support.

Key Projects:

AI Bull — AI-Powered Trading Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, and back-testing.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual responses.
- Integrated LLM capabilities into application workflows, enabling conversational access to relevant financial information.

- Built FastAPI/Python backend APIs and business logic for financial data processing, application workflows, and service integration.
- Developed back-testing engines and configurable screening and filtering capabilities for stocks and mutual funds.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested APIs and workflows with Postman and resolved bugs across CRM functionality and production workflows.

GharPe — Real Estate Management Platform

React.js, FastAPI, Python, MongoDB, REST APIs

- Developed the React.js-based admin system for managing real estate projects and controlling their visibility and availability on the website.
- Built project management workflows allowing administrators to create, update, configure, and manage projects displayed across the platform.
- Developed lead management functionality to track and manage customer inquiries generated through the platform.
- Built builder management features for maintaining builder information and associating builders with their respective projects.
- Developed and integrated FastAPI/Python REST APIs supporting administrative workflows, project configuration, and data operations.

PERSONAL PROJECTS

ML & LLM Data Intelligence Platform — Machine Learning and RAG-Based Data Intelligence Platform

Python, Pandas, NumPy, scikit-learn, FastAPI, RAG, LLM, Vector Search, AWS S3

- Built a data-processing pipeline for ingesting, cleaning, validating, transforming, and preparing structured and unstructured datasets for machine learning and LLM workflows.
- Implemented feature engineering and baseline classification and regression models using Python and scikit-learn, with model evaluation using appropriate validation metrics.
- Developed a retraining workflow to refresh models when new validated training data becomes available and track model evaluation results.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using vector search to retrieve relevant context and provide grounded LLM responses.
- Exposed data-processing, model-inference, and RAG capabilities through FastAPI services and REST APIs for production-style application integration.
- Added validation, error handling, logging, and modular service design to support reliable ML service delivery and maintainable deployment workflows.

[JS Studio](#) — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

[AnyScrum](#) — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

[Guider](#) — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

[Contact Management App](#) — Contacts Storage Application

React.js, JavaScript, Netlify

- Developed a React-based application for creating, viewing, updating, and managing contact information.
- Built reusable UI components and deployed the application on Netlify.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur